A GENERAL SHAPE AND FLUCTUATION THEOREM FOR INWARD-EXCITED WALKS

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Abstract. We study nearest-neighbor walks whose first departure from each site has an inward mean prescribed by a convex gauge, and whose later departures are simple random walk. The first-departure laws may depend on the past, and the drift strength may have spatial defects of codimension one. One potential argument gives an almost-sure filled limit shape, uniform conical local times, and recurrence. In the Euclidean case it retains the sharp planar fluctuation exponent, polylogarithmic radius bounds in higher dimensions, and the local-time limit laws. A quadratic martingale gives Gaussian and iterated-logarithm laws for a weighted radial moment for every convex body. An energy identity additionally controls the number of sites with small local time. The proof combines exit-time stopping, contact-point positivity, and a finite-scale calibration, and includes the three previously studied models.

1. The theorem and its specializations

The range of an inward-excited walk is determined at first order by its mean excitation. The probabilities realizing that mean, and spatial changes confined to sufficiently thin sets, affect only lower-order terms. We make this principle quantitative. The argument unifies Singh [6], Bou-Rabee and Peres [1], and Nguyen [5]; all three models are special cases of the same assumptions below.

Fix $d \geq 2$. Let $K \subset \mathbb{R}^d$ be compact and convex, with 0 in its interior, and let $\rho(x) = \inf\{t > 0 : x \in tK\}$ be its gauge. Write $V = |K|$ and $\omega_d = |B(0,1)|$. Symmetry and smoothness of K are not required. Fix $\kappa > 0$, subgradients $\xi(x) \in \partial\rho(x)$ for $x \in \mathbb{Z}^d \setminus \{0\}$, and weights $\lambda(x) \geq \lambda_- > 0$. Assume that, for a constant $A < \infty$,

$$\sum_{x \in \mathbb{Z}^d \cap B(y,s)} |\lambda(x) - 1| \leq A s^{d-1} \qquad (y \in \mathbb{R}^d,\ s \geq 1). \tag{1.1}$$

Set $\xi(0) = 0$ and $\lambda(0) = 1$. Taking $s = 1$ above gives $\lambda(x) \leq 1 + A$, so no separate upper bound is needed.

The walk starts at $X_0 = 0$. On its first departure from x, its conditional step distribution, given the entire past, can be any distribution on $\{\pm e_1, \dots, \pm e_d\}$ with mean

$$-\kappa\lambda(x)\xi(x). \tag{1.2}$$

On every later departure it chooses its neighbor uniformly. The first departure from zero may have any mean-zero nearest-neighbor law. Only existence of the distributions in (1.2) is assumed; in particular no lower bound on individual probabilities is required. The fields λ and ξ are deterministic. Allowing the realizing distributions to depend on the past is useful because every estimate below uses their conditional means and bounded steps.

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Put

$$\ell_n(x) = \sum_{j=0}^{n-1} \mathbf{1}_{\{X_j = x\}}, \qquad A_n = \{x : \ell_n(x) > 0\}, \qquad D_n = \bigcup_{x \in A_n} \big(x + [-1/2, 1/2)^d\big), \tag{1.3}$$

$$r_n = \left(\frac{(d+1)n}{2d\kappa V}\right)^{1/(d+1)}, \quad b_n = \inf_{y \notin D_n} \rho(y), \quad B_n = \max_{j \le n} \rho(X_j). \tag{1.4}$$

Thus $\int 2d\kappa(r_n - \rho)_+ = n$. Tildes denote constant extension of lattice functions to the unit cells. All probability estimates below concern integers $n \ge 2$. Constants may depend on $d, K, \kappa, \lambda_-, A$ and a specified probability exponent p, but are uniform over the allowed first-departure laws.

Theorem 1.1 (Shape, occupation, and moment universality)**.** *Under (1.1)–(1.2), the following hold.*

(i) *Almost surely, for every $\varepsilon > 0$ and all sufficiently large n,*

$$\{x \in \mathbb{Z}^d : \rho(x) < (1-\varepsilon)r_n\} \subset A_n \subset \{x \in \mathbb{Z}^d : \rho(x) < (1+\varepsilon)r_n\},$$

and $r_n^{-1} \max_x |\ell_n(x) - 2d\kappa(r_n - \rho(x))_+| \to 0$. In particular every site is visited infinitely often.

(ii) *Define*

$$h_n = \begin{cases} \sqrt{r_n \log n}, & d = 2, \\ \log n, & d \ge 3, \end{cases} \qquad q_n = h_n / r_n.$$

For every $p > 0$, with probability at least $1 - C_p n^{-p}$,

$$|b_n - r_n| \le C_p r_n \sqrt{q_n}, \tag{1.5}$$

$$B_n - r_n \le C_p r_n q_n^{1/(d+1)} \log n, \tag{1.6}$$

$$\max_x |\ell_n(x) - 2d\kappa(r_n - \rho(x))_+| \le C_p r_n q_n^{1/(d+1)}. \tag{1.7}$$

(iii) *Set*

$$\delta_n = \begin{cases} \sqrt{r_n} \log n, & d = 2, \\ \sqrt{r_n \log n}, & d \ge 3. \end{cases}$$

On an event of probability at least $1 - C_p n^{-p}$, simultaneously for every real $t \ge 0$,

$$\#\{x \in A_n : \ell_n(x) \le t\} \le C_p (t + \delta_n) r_n^{d-1}. \tag{1.8}$$

(iv) *Let*

$$S_n = \sum_{x \in A_n} \lambda(x)\rho(x), \qquad J_K = \int_K |z|^2 \, dz, \qquad \sigma_K^2 = \frac{2J_K}{\kappa(d+3)}.$$

Then

$$\frac{S_n - n/(2\kappa)}{r_n^{(d+3)/2}} \Longrightarrow N(0, \sigma_K^2), \tag{1.9}$$

and, for either sign, almost surely,

$$\limsup_{n \to \infty} \frac{\pm(S_n - n/(2\kappa))}{r_n^{(d+3)/2} \sqrt{2 \log\log n}} = \sigma_K. \tag{1.10}$$

For $R_{\mathrm{mom}}(n) = ((d+1)S_n/(dV))^{1/(d+1)}$, the same assertions hold with $S_n - n/(2\kappa)$ replaced by $R_{\mathrm{mom}}(n) - r_n$, the exponent $(d+3)/2$ replaced by $(3-d)/2$, and σ_K replaced by $\sigma_K/(dV)$.

The weights in the moment statistic are essential: condition (1.1) allows an $O(r_n^d)$ change in $\sum_{A_n} \rho(x)$, which can exceed its Gaussian normalization. They disappear when $\lambda \equiv 1$.

For the Euclidean case put $R_{\mathrm{in}} = b_n$, $R_{\mathrm{out}} = B_n$, and $W_n = R_{\mathrm{out}} - R_{\mathrm{in}}$. These definitions use cells for the inner radius and include X_n in the outer radius.

Theorem 1.2 (Euclidean refinement)**.** *Under the assumptions of Theorem 1.1, with $K = \{|x| \le 1\}$, all the following conclusions hold for the same class of cookie laws.*

(i) *For every $p > 0$, with probability at least $1 - C_p n^{-p}$,*

$$|R_{\mathrm{in}} - r_n| \le C_p h_n, \qquad |D_n \triangle B(0, r_n)| \le C_p r_n^{d-1} h_n, \tag{1.11}$$

$$R_{\mathrm{out}} - r_n \le C_p \begin{cases} \sqrt{r_n}(\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \ge 3, \end{cases} \tag{1.12}$$

$$\max_x |\ell_n(x) - 2d\kappa(r_n - |x|)_+| \le C_p \begin{cases} \sqrt{r_n}(\log n)^{3/2}, & d = 2, \\ \sqrt{r_n \log n}, & d \ge 3. \end{cases} \tag{1.13}$$

For each fixed $\theta < 1$, restricting the maximum to $|x| \le \theta r_n$ improves the planar bound to $C_{p,\theta}\sqrt{r_n}\log n$.

(ii) *Let G be the simple-random-walk Green function when $d \ge 3$, and put*

$$a_n = \begin{cases} \sqrt{r_n \log r_n}, & d = 2, \\ \sqrt{r_n}, & d \ge 3, \end{cases} \qquad \Sigma(y,z) = \begin{cases} 16\kappa/\pi, & d = 2, \\ 2d\kappa(2G(y-z) - \mathbf{1}_{\{y=z\}}), & d \ge 3. \end{cases}$$

For any finite set of fixed sites, the vector $a_n^{-1}(\ell_n(y) - 2d\kappa(r_n - |y|))$ converges in distribution to a centered Gaussian vector with covariance Σ. At each fixed site and for either sign, almost surely,

$$\limsup_n \frac{\pm(\ell_n(y) - 2d\kappa(r_n - |y|))}{a_n\sqrt{2\log\log n}} = \sqrt{\Sigma(y,y)}. \tag{1.14}$$

(iii) *In dimension two, for every $p > 0$ there is $c_p > 0$ such that each of*

$$\mathbb{P}\{r_n - R_{\mathrm{in}} \ge c_p\sqrt{r_n \log n}\}, \quad \mathbb{P}\{R_{\mathrm{out}} - r_n \ge c_p\sqrt{r_n \log n}\}, \quad \mathbb{P}\{W_n \ge c_p\sqrt{r_n \log n}\} \tag{1.15}$$

is at least n^{-p} for all large n. Almost surely,

$$\limsup_n \frac{r_n - R_{\mathrm{in}}}{\sqrt{r_n \log\log n}}, \quad \limsup_n \frac{R_{\mathrm{out}} - r_n}{\sqrt{r_n \log\log n}} \ge \frac{1}{\sqrt{10\pi\kappa}}, \tag{1.16}$$

$$\limsup_n \frac{W_n}{\sqrt{r_n \log\log n}} \ge \sqrt{\frac{\pi}{3\kappa}}. \tag{1.17}$$

For every $a \ge 0$,

$$\limsup_n \mathbb{P}\{W_n \le a\sqrt{r_n}\} \le 1 - \exp(-3\kappa a^2/\pi). \tag{1.18}$$

(iv) *In every dimension, for some $c > 0$ and all large n,*

$$\mathbb{P}\left\{ \max_{|x| \le \sqrt{r_n}} |\ell_n(x) - 2d\kappa(r_n - |x|)| \ge c \begin{cases} \sqrt{r_n}\log n, & d = 2, \\ \sqrt{r_n \log n}, & d \ge 3 \end{cases} \right\} \ge 1 - n^{-c}. \tag{1.19}$$

The bounds in part (i) *hold eventually almost surely.*

We also retain the logarithmic radial lower bounds under the additional assumption of a uniform positive lower bound on all first-departure probabilities; their full tail forms are (7.14)–(7.15) below. The proof yields two additional boundary estimates. Write $R = R_{\mathrm{out}}(n) + 2\sqrt{d}$ and $E_n = D_n \setminus B(0, R_{\mathrm{in}}(n))$. With the same probability as in Theorem 1.2, uniformly over unit vectors u and $t \ge 1$,

$$|E_n \cap B(Ru, t)| \le C_p h_n t^{d-1}. \tag{1.20}$$

For lattice sites with $R_{\text{in}} \le |y| \le R$, put $s = R - |y| + 1$. Then

$$\ell_n(y) \le C_p \begin{cases} h_n + \sqrt{h_n s}, & d = 2, \\ \log n + ((\log n) s^{d-1})^{1/d}, & d \ge 3. \end{cases} \tag{1.21}$$

Recovery of the three models. For the radial model of Singh [6], take $K = \{|x| \le 1\}$, $\lambda \equiv 1$, and $\kappa = \beta$. His prescribed-mean assumption is exactly (1.2). For the norm model of Bou-Rabee and Peres [1], take $\lambda \equiv 1$ and the particular law $q_x(\pm e_i) = 1/(2d) \mp \kappa \xi_i(x)/2$. Their strict parameter restriction guarantees existence and, for the logarithmic lower bounds, uniform ellipticity. Theorem 1.2 retains the Euclidean constants and exponents, and $J_{B(0,1)} = d\omega_d/(d+2)$ recovers their moment variances.

For Nguyen [5], let $K = \{\|x\|_1 \le 1\}$, $\xi(x) = \operatorname{sgn}(x)$, $\kappa = \theta/d$, and $\lambda(x) = d/m(x)$ at $x \ne 0$, where $m(x)$ is the number of its nonzero coordinates. Use his first-departure kernel. The weights differ from one only on the union of the coordinate hyperplanes, which contains at most Cs^{d-1} lattice points in any ball of radius $s \ge 1$. Thus (1.1) holds, including at $\theta = 1$. Theorem 1.1 strengthens Hausdorff convergence in probability to almost-sure filled inclusions and uniform local-time convergence. Since $|K| = 2^d/d!$, its radius is $r_n = \frac{1}{2}((d+1)!n/\theta)^{1/(d+1)}$, with the original normalization. This verifies model containment at the transition-rule level, including the exceptional coordinate hyperplanes. The proof. Green martingales express local times as the potential of the occupied set with a small error. Singh's exit-time stopping gives the initial occupation bound. At the first hole, convexity makes every excess contribution nonnegative. This contact estimate and a single exponential crossing bound give the shape. Bou-Rabee and Peres's contact and exterior-probe method gives the fluctuation bounds. A finite-scale form of Nguyen's energy calibration supplies (1.8) directly. Once the cone is known, later departures dominate all relevant quadratic variations, explaining the universality of the Gaussian constants. We give the moderate-deviation argument separately because a central limit theorem alone does not imply the lower tails.

2. A common potential argument

We use the assumptions and notation of Section 1. Put $c = \min_{|u|=1} \rho(u) > 0$, $\Lambda = \max_{|u|=1} \rho(u)$, and $C_x = x + [-1/2, 1/2)^d$. Write q_x for the conditional law used at the first departure from x; this law can be random, but its prescribed mean is fixed. Every estimate below is pathwise uniform over its possible values. Set $\Delta = P - I$ for the simple-random-walk generator and $\bar{\nabla}_i f(x) = (f(x+e_i) - f(x-e_i))/2$. For a bounded measurable D, define

$$U_D(y) = \frac{2\kappa}{\omega_d} \int_D \nabla\rho(v) \cdot \frac{v-y}{|v-y|^d}\, dv. \tag{2.1}$$

Lemma 2.1 (Cone, layers, and contact)**.** *For $b \ge 0$, $U_{bK}(y) = 2d\kappa(b - \rho(y))_+$. If $E \subset \{b \le \rho \le b + a\}$ then $\|U_E\|_\infty \le 2d\kappa a$; for arbitrary E, $\|U_E\|_\infty \le C|E|^{1/d}$. Suppose $bK \subset D \subset B(0,R)$ up to null sets, $y_0 \in \partial(bK)$ lies in $\overline{D^c}$, and $E = D \setminus bK$. Put $H = U_D(y_0)$ and $I = \int_E (\rho - b)$. Then*

$$H \ge 0, \qquad I \le CR^d H, \qquad \|U_E\|_\infty \le CI^{1/(d+1)}. \tag{2.2}$$

Proof. For a compactly supported Lipschitz f, polar coordinates about y give $\int \nabla f(v) \cdot (v - y)|v - y|^{-d}\, dv = -d\omega_d f(y)$. Apply this to $f = (b - \rho)_+$. On each ray $y + tu$, convexity makes $t \mapsto \rho(y + tu)$ decrease and then increase. On either part, its total variation while $b \le \rho \le b + a$ is at most a; the two signed parts therefore give an integral between $-a$ and a. Integrating over directions proves the layer bound. Rearrangement of $|v - y|^{1-d}$ proves the volume bound. Finally $\nabla\rho(v) \cdot (v - y_0) \ge \rho(v) - \rho(y_0) = \rho(v) - b$ on E, proving the first two assertions in (2.2). Split E at $\rho = b + a$: the layer contributes at most Ca and the rest, of volume at most I/a, contributes at most $C(I/a)^{1/d}$. Take $a = I^{1/(d+1)}$; the case $I = 0$ follows directly. □

Lemma 2.2 (Discrete potential representation)**.** *Let g be the potential kernel in dimension two and minus the Green function in dimensions at least three, normalized by $\Delta g = \mathbf{1}_{\{0\}}$, where $\Delta = P - I$*

for simple random walk. Write $g_y(x) = g(x-y)$ *and* $M^y = M^{g_y}$ *for its Dynkin martingale; for finite combinations of these kernels, extend* M^f *linearly. These martingales have bounded increments. Whenever* $\max_{j\le t}|X_j| \le R$, $R \ge 2$, *and* $y \in \mathbb{Z}^d$ *satisfies* $|y| \le R^{d+2}$,

$$\ell_t(y) = U_{D_t}(y) - M_t^y + O(\log R), \qquad \langle M^y\rangle_t \le CW_t(y), \quad W_t(y) = \sum_x \ell_t(x)(1+|x-y|)^{2-2d}. \tag{2.3}$$

Moreover, for $|z-y| \le \sqrt{d}$, $|U_{D_t}(z) - U_{D_t}(y)| \le C\log R$.

Proof. Standard Green estimates [4] give

$$\bar\nabla g(z) = \frac{2}{\omega_d}\frac{z}{|z|^d} + O((1+|z|)^{-d}) \quad (z \ne 0),$$
$$|g(z+e) - g(z)| \le C(1+|z|)^{1-d},$$
$$|g(z+e_i) + g(z-e_i) - 2g(z)| \le C(1+|z|)^{-d}.$$

Write $s_i(x) = q_x(e_i) + q_x(-e_i)$. Exactly,

$$(Q_x - P)g_y(x) = -\kappa\lambda(x)\xi(x)\cdot\bar\nabla g(x-y) + \frac{1}{2}\sum_i (s_i(x) - 1/d)[g(x+e_i-y) + g(x-e_i-y) - 2g(x-y)].$$

Dynkin's formula, grouping first departures by site, gives the first identity in (2.3) with the discrete drift potential. The displayed second differences sum to $O(\log R)$. The endpoint $g(X_t - y) - g(-y)$ has this bound too. By dyadic annuli,

$$\sup_y \sum_{x\in B(0,CR)} |\lambda(x) - 1|(1+|x-y|)^{1-d} \le C\log R. \tag{2.4}$$

Thus replacing λ by one costs $O(\log R)$, even for an arbitrary subset of visited defect sites. Replacing the Green gradient by its leading term, and then by cell averages, has the same cost.

For completeness the subgradient replacement also costs only $O(\log R)$, including at corners of K. The positive measure $\mu = \Delta_{\mathrm{cont}}\rho$ satisfies $\mu(B(y,s)) \le Cs^{d-1}$, by testing against a cutoff and using $|\nabla\rho| \le \Lambda$. The convex-cell estimate is

$$\int_{C_x} |\nabla\rho(v) - \xi(x)|\,dv \le C\mu(B(x, 6\sqrt{d})). \tag{2.5}$$

Indeed, $h(v) = \rho(v) - \rho(x) - \xi(x)\cdot(v-x)$ is nonnegative, convex, and zero at x. Convexity along coordinate lines gives $|\partial_i h(v)| \le h(v+e_i) + h(v-e_i)$, so the left side is at most $C\int_{B(x,2\sqrt{d})} h$. Mollify h and put $A = 2\sqrt{d}$. Convexity gives

$$h_\varepsilon(x+tu) \le h_\varepsilon(x) + t\,\partial_r h_\varepsilon(x+2Au) \qquad (0 \le t \le A,\ |u| = 1).$$

Integration in t, u and the divergence theorem bound the integral by $Ch_\varepsilon(x) + C\mu(B(x,6\sqrt{d}))$. Letting $\varepsilon \downarrow 0$ proves (2.5). Bound the cells close to y directly; bounded overlap and dyadic annuli bound the remaining error by $C\int_{1\le|z-y|\le CR^{d+2}}(1+|z-y|)^{1-d}\,d\mu(z) = O(\log R)$. The increment estimate gives the bracket bound. Splitting the integral at distance $2\sqrt{d}$ and differentiating the far kernel gives the cell modulus. □

Lemma 2.3 (Exit bootstrap)**.** *Put* $\tau_R = \inf\{t : \rho(X_t) \ge R\}$. *For every* $p > 0$, *outside an event of probability* CR^{-p}, *simultaneously for* $t \le \tau_R$,

$$\max_x \ell_t(x) \le CR, \qquad \tau_R \le CR^{d+1}, \qquad \sup_{|y|\le R^{d+2}} |M_t^y| \le C\sqrt{R}\log R. \tag{2.6}$$

Consequently $\|\widetilde{\ell}_t - U_{D_t}\|_\infty \le C\sqrt{R}\log R$ *on* $B(0,CR)$.

Proof. To see that the exit time is finite, stop the compensated coordinate martingale $Z_t^i = X_t^i + \kappa \sum_{x \in A_t} \lambda(x)\xi_i(x)$ at τ_R. It is bounded by a deterministic constant C_R. There are at most $m_R = O(R^d)$ first departures before exit, so its bracket is at least $((t \wedge \tau_R) - m_R)/d$. Consequently $\mathbb{E}(t \wedge \tau_R) \le m_R + dC_R^2$, and $\tau_R < \infty$ almost surely. Before exit, all departed sites lie in $B(0, CR)$. The exact discrete Dynkin identity and the first Green increment bound give $\ell_t(y) \le |M_t^y| + CR$ without any preliminary clock estimate. Stop also when one of the displayed martingales first reaches R in absolute value. Up to this stopping time, $\max_x \ell(x) \le C'R$, and hence the clock is at most $C'R^{d+1}$ and every bracket at most $C'R \log R$. Freedman's maximal inequality [2, Theorem 1.6] and a union bound over the polynomially many poles give $\sup |M^y| \le C_p \sqrt{R} \log R$ with probability $1 - CR^{-p}$. This is smaller than R for large R, so the additional stop is never reached before τ_R. The representation and cell modulus finish. □

Lemma 2.4 (Exponential crossing and projection)**.** *For every $p > 0$ and integer $N \ge 2$, outside an event of probability CN^{-p}, simultaneously for $t \le N$ and integers $0 \le k \le C_0 N$,*

$$\max_{j \le t} \operatorname{dist}(X_j, kK) \le C \left(1 + \max_{\rho(x) \ge k} \ell_t(x)\right) \log N. \tag{2.7}$$

Here C_0 is any fixed constant.

Proof. We first record the short crossing estimate. Fix a unit vector u, $a \ge 0$, and suppose all departed sites in the cap $u \cdot x > a$ have $u \cdot \lambda(x)\xi(x) \ge \alpha > 0$. With $f(x) = e^{\theta u \cdot x}$, conditional increments satisfy

$$\mathbb{E}[f(X_{j+1}) - f(X_j) \mid \mathcal{F}_j] \le f(X_j)(C\theta^2 - \kappa\alpha\theta I_j) \quad (u \cdot X_j > a),$$

where I_j indicates a first departure. Stop at time N, on entering a cap site violating the drift condition, when a cap local time exceeds L, or when $u \cdot X$ reaches $a + h$. Choose $\theta = c/(L+1)$. Summing by site makes the total contribution from the cap nonpositive: each departed site supplies its cookie once and at most $L + 1$ visits. The other at most N increments contribute at most $CNe^{\theta(a+1)}$. Optional stopping of the Dynkin martingale gives

$$\mathbb{P}\left(\begin{array}{c} \text{the level } a+h \text{ is reached by } N, \text{ all previously visited} \\ \text{cap sites satisfy the drift condition, and cap local times} \le L \end{array} \right) \le CN \exp\{-ch/(L+1)\}. \tag{2.8}$$

The stopping argument does not condition on the event in this display.

Let $w = \max_{j \le t} \operatorname{dist}(X_j, kK)$ and choose $x_* = X_j$ attaining it. When $w > 0$, let π denote Euclidean projection onto kK, $u = (x_* - \pi(x_*))/w$, and $\beta = u \cdot \pi(x_*)$. Then $u \cdot X_j \le T := \beta + w$ for all $j \le t$ and $\beta \ge 0$. Put $\eta = c^2/(8\Lambda^2)$. If x is a site visited by time t and $u \cdot x > T - \eta w$, write $v = (x - \pi(x))/|x - \pi(x)|$. The projection inequalities give $u \cdot v > 1 - \eta$; moreover

$$\xi(x) \cdot v \ge \frac{\rho(x) - k}{\operatorname{dist}(x, kK)} \ge c,$$

where the last inequality follows by comparing $\pi(x)$ with $kx/\rho(x)$. Thus $u \cdot \lambda(x)\xi(x) \ge \lambda_- c/2$. All these cap sites satisfy $\rho(x) > k$. Use (2.8) with a the least integer at least $T - \eta w$, and with L the integer above the largest relevant local time. The directions u belong to the deterministic list formed by projecting the lattice sites $|x| \le N$ onto kK, for integer $k \le C_0 N$. This list, the integer levels $a \le CN$, and $1 \le L \le N + 1$ have polynomial total size. For each triple choose the deterministic length $C_p(L+1)\log N$ in (2.8). A union bound excludes all these crossings. The drift condition is needed only at visited sites, since the stopped estimate ends on entering any violating site. Since $T - a \ge \eta w - 1$, this proves (2.7); bounded w is absorbed into the constant. □

Proposition 2.5 (Coarse scale, with polynomial tails)**.** *For every $p > 0$ there are $c_p, C_p > 0$ such that, with probability at least $1 - C_p R^{-p}$,*

$$c_p R^{d+1} \le \tau_R \le C_p R^{d+1}. \tag{2.9}$$

Consequently, with $r_n = ((d+1)n/(2d\kappa|K|))^{1/(d+1)}$, *with probability at least* $1 - C_p n^{-p}$,

$$\max_{j\le n}|X_j| \le C_p r_n, \quad c_p r_n \le \max_x \ell_n(x) \le C_p r_n, \quad c_p r_n^d \le |A_n| \le C_p r_n^d. \tag{2.10}$$

Proof. Work on the events of the preceding two lemmas with a deterministic horizon $N = \lceil CR^{d+1}\rceil$. At time τ_R, let $b = \inf_{y\notin D_{\tau_R}} \rho(y)$, and choose a contact point $y_0 \in \partial(bK) \cap \overline{D^c_{\tau_R}}$ in the closure of an unoccupied cell. Continuity and the cell estimate imply $0 \le H = U_{D_{\tau_R}}(y_0) \le \delta_R = C\sqrt{R}\log R$. Lemma 2.1 therefore gives

$$\|\widetilde{\ell}_{\tau_R} - 2d\kappa(b-\rho)_+\|_\infty \le \delta_R + C(R^d\delta_R)^{1/(d+1)} =: \varepsilon_R = o(R/\log R).$$

Use (2.7) with $k = \lfloor b\rfloor$; outside kK the local times are at most $C + \varepsilon_R$. Since $\rho(X_{\tau_R}) \ge R$ and ρ is Λ-Lipschitz,

$$R - b \le C + C(\varepsilon_R + 1)\log R = o(R).$$

The cone is supported on $\{\rho < b\} \subset D_{\tau_R}$. Integrating its approximation over $D_{\tau_R} \subset B(0, CR)$ gives

$$\tau_R = \frac{2d\kappa|K|}{d+1} b^{d+1} + O(R^d\varepsilon_R) \asymp R^{d+1},$$

proving (2.9). To obtain (2.10), choose $R = \lceil Cn^{1/(d+1)}\rceil$ so that the lower clock bound exceeds n, and apply the exit event with exponent $(d+1)(p+1)$. Then $n < \tau_R$, so the spatial and local-time upper bounds follow. The remaining bounds follow from $n = \sum_x \ell_n(x) \le |A_n| \max_x \ell_n(x)$ and $|A_n| \le C(1 + \max_{j\le n}|X_j|)^d$. □

3. Quantitative stability of the cone

Fix $p > 0$, write $r = r_n$, $a = \log n$, and work on one event of probability at least $1 - Cn^{-p}$. Proposition 2.5, Lemma 2.2, Freedman's inequality and a union bound give, simultaneously for all lattice sites $|y| \le 3n$,

$$D_n \subset B(0, Cr), \quad \max \ell_n \le Cr, \quad |M_n^y| \le C(\sqrt{aW_n(y)} + a), \quad \ell_n(y) = U_{D_n}(y) - M_n^y + O(a). \tag{3.1}$$

Here and below the constants may be increased by increasing the exponent in the constituent probability estimates. Cell interpolation gives

$$\|\ell_n - U_{D_n}\|_\infty \le \delta, \qquad \delta = \begin{cases} C\sqrt{r}\,a, & d = 2, \\ C\sqrt{ra}, & d \ge 3. \end{cases} \tag{3.2}$$

For points outside $B(0, 2n)$ this follows directly from the decay of the potential, since $\ell_n = 0$ there. Define

$$b = \inf_{y\notin D_n} \rho(y), \quad E = D_n \setminus bK, \quad H = U_{D_n}(y_0), \quad I = \int_E (\rho(v) - b)\,dv,$$

where $y_0 \in \partial(bK)$ is in the closure of an empty cell C_z. Thus $\ell_n(z) = 0$, $|z - y_0| \le \sqrt{d}/2$, and convexity gives $H = U_E(y_0) \ge 0$. Set

$$h = \begin{cases} \sqrt{ra}, & d = 2, \\ a, & d \ge 3, \end{cases} \qquad q = h/r. \tag{3.3}$$

Lemma 3.1 (Noise at the first hole)**.** *On the preceding event,* $H \le Ch$.

Proof. The equation at the empty site and the cell modulus give

$$H \le C(\sqrt{aW_n(z)} + a). \tag{3.4}$$

For every nonnegative integrable radial function χ, Newton's shell identity gives, with $m(s) = \int_{|v|<s} \chi(v)\,dv$,

$$(\chi * U_D)(y) = \frac{2\kappa}{\omega_d} \int_D \nabla\rho(v) \cdot \frac{v-y}{|v-y|^d} m(|v-y|)\,dv. \tag{3.5}$$

Indeed the spherical average of $(v-y-w)/|v-y-w|^d$ on $|w| = s$ is $(v-y)|v-y|^{-d}\mathbf{1}_{\{|v-y|>s\}}$; integrate in s. Since the contact integrand is nonnegative, $(\chi * U_E)(y_0) \le \|\chi\|_1 H$.

In the plane take $\chi(v) = (1+|v|)^{-2}\mathbf{1}_{\{|v|\le Cr\}}$, with the cutoff large enough to cover $D_n - y_0$. Cell comparison and (3.2) give

$$W_n(z) \le C(\chi * \ell_n)(y_0) \le C(\chi * U_{bK})(y_0) + C(H+\delta)\|\chi\|_1.$$

The cone $U_{bK} = 2d\kappa(b-\rho)_+$ is Lipschitz and vanishes at y_0, so its convolution at y_0 is $O(r)$. Also $\|\chi\|_1 = O(a)$ and $H \le \delta$ by (3.2). Consequently $W_n(z) \le C(r + \sqrt{r}\,a^2) \le Cr$, proving the claim.

For $d \ge 3$ put $\chi(v) = (1+|v|)^{2-2d}$. This kernel is integrable and $\chi * \chi \le A\chi$: split the convolution into $|w| \ge |v|/2$ and $|v-w| \ge |v|/2$. Cell comparison, the martingale bound and Young's inequality yield, for every fixed $\gamma > 0$,

$$\ell_n \le U_{D_n} + C_\gamma a + \gamma\chi * \ell_n.$$

This inequality holds globally; outside $B(0,2n)$ the local time is zero and $|U_{D_n}| \le C$. Convolve it with χ and take $\gamma = (2A)^{-1}$ to obtain the resolvent estimate

$$\chi * \ell_n \le 2\chi * U_{D_n} + Ca. \tag{3.6}$$

The cone is Lipschitz, bounded by Cb, and $b \le Cr$, hence

$$(\chi * U_{bK})(y) \le \|\chi\|_1 U_{bK}(y) + Ca; \tag{3.7}$$

the error is bounded by $C\int \chi(v)\min\{|v|, r\}\,dv = O(a)$. At y_0, (3.5) and (3.6) now give $W_n(z) \le C(\chi * \ell_n)(y_0) \le C(H+a)$. Substitute in (3.4) and absorb $C\sqrt{aH} \le H/2 + Ca$. □

Rates for every convex body. Lemma 2.1 and the contact bound give $I \le Cr^d h = Cr^{d+1}q$. Split E into the layer $b \le \rho < b+s$, whose volume is at most $Cr^{d-1}s$, and the remainder, whose volume is at most I/s. With $s = r\sqrt{q}$ this proves

$$|E| \le Cr^d\sqrt{q}, \qquad \|U_E\|_\infty \le Crq^{1/(d+1)}. \tag{3.8}$$

The quadratic martingale is

$$Q_n = |X_n|^2 - n + 2\kappa \sum_{x\in A_n} \lambda(x)\rho(x).$$

Stopping on leaving $B(0,Cr)$ and applying Azuma gives $|Q_n| \le Cr\sqrt{na}$. The sparse-weight condition and cell quadrature each cost $O(r^d)$ in replacing the weighted sum by $\int_{D_n}\rho$. Since $\int_{bK}\rho = d|K|b^{d+1}/(d+1)$ and $n = 2d\kappa|K|r^{d+1}/(d+1)$, we obtain

$$\left|1 - (b/r)^{d+1}\right| \le C\left(r^{(1-d)/2}\sqrt{a} + r^{-1} + |E|r^{-d}\right). \tag{3.9}$$

The first two terms are at most $C\sqrt{q}$; hence

$$|b-r| \le Cr\sqrt{q}, \qquad \max_x |\ell_n(x) - 2d\kappa(r-\rho(x))_+| \le Crq^{1/(d+1)}. \tag{3.10}$$

Here we used the cone identity, (3.2) and (3.8); their errors are bounded by the right side. Apply the projection crossing estimate (2.7) with $k = \lfloor r \rfloor$. Outside kK the local time is at most $C(1+rq^{1/(d+1)})$. Since ρ is Lipschitz, it follows that

$$\max_{j\le n} \rho(X_j) \le r + Crq^{1/(d+1)}a. \tag{3.11}$$

These prove Theorem 1.1(ii). Taking $p > 1$ and applying Borel–Cantelli gives the filled inclusions and uniform cone law in part (i); at each fixed site the cone grows linearly in r_n, proving recurrence.

4. Euclidean geometry and exterior probes

Now let $\rho(v) = |v|$. We retain r, a, b, E, h, q above, so $b = R_{\mathrm{in}}(n)$. For $|v| \ge b = |y_0|$ the exact identity

$$2(|v|^2 - v \cdot y_0) = |v - y_0|^2 + |v|^2 - b^2$$

gives

$$u_v \cdot \frac{v - y_0}{|v - y_0|^d} \ge \frac{|v - y_0|^{2-d}}{2|v|} \ge c r^{1-d} \qquad (v \in E).$$

Thus the contact estimate improves the volume and inner radius bounds to

$$|E| \le C r^{d-1} h = C r^d q, \quad |b - r| \le C r q, \quad |D_n \triangle B(0, r)| \le C r^d q. \tag{4.1}$$

The radius assertion follows from (3.9) since its first two terms are at most Cq.

Write U_E^+ for the radial potential with its negative integrand replaced by zero, and put

$$L = 1 + \max_{|x| \ge b} \ell_n(x), \qquad w = R_{\mathrm{out}}(n) - b + 2\sqrt{d}, \qquad R = b + w.$$

Then $w \ge 1$ and $E \subset \{b \le |v| \le R\}$. Crossing with $k = \lceil b \rceil$ gives $w \le CLa$. For $d \ge 3$, (3.5) gives

$$\chi * U_E \le \|\chi\|_1 U_E^+, \qquad \chi * U_E \le \|\chi\|_1 \|U_E\|_\infty.$$

Combining these with the resolvent and cone-convolution estimates yields

$$\ell_n(y) \le C\big((b - |y|)_+ + \|U_E\|_\infty + a\big), \quad \ell_n(y) \le C(U_E^+(y) + a) \quad (|y| \ge b). \tag{4.2}$$

In the plane use (3.2) instead. In either case the volume bound $\|U_E\|_\infty \le C|E|^{1/d}$ gives

$$L \le C r q^{1/d}, \qquad w \le C r q^{1/d} a = o(r a^{-d}). \tag{4.3}$$

In particular $b \sim r$ and $w = o(r)$. The layer estimate also gives $\|U_E\|_\infty \le Cw$.

One exterior estimate in all dimensions. Set

$$\Theta = \begin{cases} h, & d = 2, \\ a + r^{-1} \sup_y \displaystyle\int_E |v - y|^{2-d}\, dv, & d \ge 3. \end{cases} \tag{4.4}$$

In either case $\Theta \ge a$ for large n and $\sup_y \int_E |v - y|^{2-d}\, dv \le Cr\Theta$; in the plane this is just $|E| \le Crh$. For $|y| \ge R$, the Euclidean identity therefore gives $U_E^+(y) \le Cr^{-1} \int_E |v - y|^{2-d}\, dv \le C\Theta$.

The planar substitute for the resolvent is particularly short. The layer estimate and (3.2) give $\ell_n(x) \le 4\kappa(b - |x|)_+ + C(w + \sqrt{r}\, a)$. For every lattice y with $|y| \ge b$, inserting this in $W_n(y)$ gives

$$W_n(y) \le Cr + C(w + \sqrt{r}\, a)a \le Cr.$$

Indeed $(b - |x|)_+ \le |x - y|$ bounds the cone contribution by Cr, and (4.3) makes the other term $o(r)$. Thus the martingale bound, and decay beyond the pole set, imply

$$|\ell_n(y) - U_E(y)| \le Ch \qquad (d = 2,\ |y| \ge b). \tag{4.5}$$

Together with (4.2), this gives in every dimension $\ell_n(y) \le C(U_E^+(y) + \Theta)$ outside the inner ball.

We now use the same exterior probe in every dimension. Given a unit vector u and $\sqrt{d} \le t \le r/4$, put $y_t = (R + 4t)u$ and let z be its cell center. Since $R = R_{\mathrm{out}} + 2\sqrt{d}$, this cell is empty and $|z| \ge R$. If $d \ge 3$, the resolvent, cone-convolution estimate and $\chi * U_E \le \|\chi\|_1 U_E^+$ give $W_n(z) \le C\Theta$; if $d = 2$, use (4.5). The martingale estimate and cell modulus give in both cases

$$|U_E(y_t)| \le C\Theta. \tag{4.6}$$

For $v \in E \cap B(Ru,t)$ we have $u_v \cdot u \geq 1 - t/R$ and

$$u_v \cdot (v - y_t) \leq R - (R+4t)(1-t/R) \leq -t, \qquad |v - y_t| \leq 5t,$$

because $t \leq R/2$ for large n. Such points contribute at most $-ct^{1-d}$ each, whereas all positive contributions total at most $C\Theta$. Hence (4.6) implies

$$|E \cap B(Ru,t)| \leq C\Theta t^{d-1} \qquad (|u| = 1,\ t \geq 1). \tag{4.7}$$

For $t > r/4$ use $|E| \leq Cr^{d-1}h \leq Cr^{d-1}\Theta$; for $1 \leq t < \sqrt{d}$ use the ball-volume bound. Thus the estimate holds down to a fixed spatial scale, independently of the shell width.

Lemma 4.1 (Cap volume controls the boundary profile)**.** *For every y with $b \leq |y| \leq R$, put $s = R - |y| + 1$. Then*

$$U_E^+(y) \leq C(\Theta s^{d-1})^{1/d} + C\Theta.$$

The same bound holds for $\ell_n(y)$ at lattice sites.

Proof. For $t \geq s$, $B(y,t) \subset B(Ru_y, 2t)$, so (4.7) gives $|E \cap B(y,t)| \leq C\Theta t^{d-1}$. The part $|v-y| \leq s$ has volume at most $C\Theta s^{d-1}$ and therefore potential at most $C(\Theta s^{d-1})^{1/d}$. For the rest, $|v|^2 - |y|^2 \leq 2Rs$ and $|v| \geq b \asymp r$ give

$$\left[u_v \cdot \frac{v-y}{|v-y|^d}\right]_+ \leq \frac{Cs}{|v-y|^d} + \frac{C}{r|v-y|^{d-2}}.$$

The cap bound makes the first integral at most $Cs\int_s^\infty \Theta t^{d-1} t^{-d-1}\,dt \leq C\Theta$; the definition of Θ bounds the second by $C\Theta$. The exterior local-time bound proves the last assertion. □

Since $s \leq w + 1 \leq 2w$, the lemma and crossing give

$$L \leq C\big((\Theta w^{d-1})^{1/d} + \Theta\big), \qquad w \leq CLa \leq \tfrac{1}{2}w + C\Theta a^d. \tag{4.8}$$

Thus $w \leq C\Theta a^d$ in every dimension. In the plane $\Theta = h$, so the lemma immediately yields

$$w \leq C\sqrt{r}\, a^{5/2}, \qquad \sup_{b \leq |y| \leq R} U_E^+(y) \leq C\sqrt{r}\, a^{3/2}. \tag{4.9}$$

For $d \geq 3$ it remains only to absorb the curvature term in Θ. If $t \geq w$ and $B(y,t)$ meets E at v, its intersection with E lies in $B(v,2t)$ and hence in a cap of radius at most $3t$. Consequently

$$|E \cap B(y,t)| \leq C\min\{\Theta t^{d-1}, r^{d-1}a\} \quad (t \geq w), \qquad |E \cap B(y,t)| \leq \omega_d t^d \quad (t < w).$$

Layer-cake integration, splitting the minimum at $t = r(a/\Theta)^{1/(d-1)}$, gives

$$\sup_y \int_E |v-y|^{2-d}\,dv \leq Cw^2 + Cr\Theta^{(d-2)/(d-1)}a^{1/(d-1)}.$$

Substitution in (4.4) and Young's inequality give

$$\Theta \leq C(a + w^2/r). \tag{4.10}$$

Together with $w \leq C\Theta a^d$, this implies $w \leq Ca^{d+1} + Ca^d w^2/r$. The last term is at most $w/2$ for large n by (4.3). We conclude that

$$w \leq Ca^{d+1}, \qquad \Theta \leq Ca. \tag{4.11}$$

Substitution in (4.7) proves the all-scale cap estimates. Lemma 4.1 now also gives

$$\ell_n(y) \leq \begin{cases} C\big(h + \sqrt{hs}\big), & d = 2, \\ C\big(a + (as^{d-1})^{1/d}\big), & d \geq 3, \end{cases} \qquad b \leq |y| \leq R, \quad s = R - |y| + 1. \tag{4.12}$$

In particular, for $d \geq 3$ the outermost radial layer of width $C_0 \log n$ has local times $O(\log n)$. If this layer reaches inside $B(0,b)$, then $w = O(a)$. The continuum boundary bound from the lemma and the maximum principle give $U_E \leq Ca$ inside that ball. There $\chi * U_E \leq \|\chi\|_1 U_E$, since the potential integrand is nonnegative; the resolvent and the cone depth $O(a)$ give the same local-time bound.

Uniform and interior local times. For $d \geq 3$, the layer bound and (4.11) give $\|U_E\|_\infty \leq Ca^{d+1}$. Together with the cone identity, (3.2) and (4.1), this gives

$$\max_x |\ell_n(x) - 2d\kappa(r - |x|)_+| \leq C\sqrt{ra}.$$

In the plane U_E is nonnegative and harmonic on $B(0,b)$, and continuous on its closure. The maximum principle and (4.9) therefore bound it throughout that ball by $C\sqrt{r}\,a^{3/2}$. On the outer annulus use (4.5); beyond the range both the local time and, except for a radial error $C|b-r|$, the target cone vanish. Consequently the uniform planar error is $C\sqrt{r}\,a^{3/2}$.

Finally, for any fixed $\theta < 1$ and $|y| \leq \theta r$, the excess set is at distance at least $c_\theta r$ for large n. Thus

$$|U_{D_n}(y) - 2d\kappa(r - |y|)| \leq C_\theta(r^{1-d}|E| + |b-r|) \leq C_\theta rq. \tag{4.13}$$

The corresponding local-time error is at most $C_\theta\sqrt{r}\,a$ in the plane and $C_\theta\sqrt{ra}$ in higher dimensions. All assertions hold simultaneously with probability $1 - Cn^{-p}$ and, by Borel–Cantelli, eventually almost surely. The all-scale cap bounds (4.7) and boundary profile (4.12) are additional quantitative conclusions of the same proof.

5. Calibration and small occupation layers

Lemma 5.1 (Finite-scale calibration)**.** *Let $a = 2d\kappa$, $b = \nabla\rho$, and let $D \subset B(0,R)$ be measurable, $R \geq 1$. Put $u = U_D$ and suppose*

$$u \geq -\delta \quad \text{on } B(0,2R), \qquad |u| \leq \delta \quad \text{on } B(0,2R) \setminus D.$$

Then $w = (u-\delta)_+$ vanishes off D, belongs to $H^1(\mathbb{R}^d)$, and, with $A = \{u > \delta\}$,

$$\int_{\mathbb{R}^d} |\nabla w + a\mathbf{1}_A b|^2 + a^2 \int_{D\setminus A} |b|^2 \leq C\delta R^{d-1}. \tag{5.1}$$

In particular $|D \setminus A| \leq C\delta R^{d-1}$.

Proof. The potential is continuous, tends to zero at infinity, and is harmonic off $\overline{D}$. The maximum principle extends $|u| \leq \delta$ from the annulus to the exterior of $B(0,R)$. The potential equation is $\operatorname{div}(\nabla u + a\mathbf{1}_D b) = 0$. Since $\mathbf{1}_D b \in L^2$, the usual Fourier multiplier estimate gives $\nabla u \in L^2$; hence the compactly supported truncation w is in H^1. Testing the equation with w, using $A \subset D$, gives

$$\int |\nabla w|^2 = -a \int b \cdot \nabla w. \tag{5.2}$$

Choose a smooth χ equal to one on $B(0,R)$, zero off $B(0,2R)$, with $|\nabla\chi| \leq C/R$ and $|\Delta\chi| \leq C/R^2$. Put $v = u - w$, so $|v| \leq \delta$. The equation tested with $\chi\rho$, together with (5.2), gives the exact identity

$$\int |\nabla w + a\mathbf{1}_D b|^2 = a\int b\cdot\nabla w + a^2\int_D |b|^2 = -a\int \nabla v\cdot\nabla(\chi\rho) = a\int v\, d\Delta(\chi\rho).$$

The convexity measure $\mu = \Delta\rho$ has $\mu(B(0,2R)) \leq CR^{d-1}$. Since $\Delta(\chi\rho) = \chi\mu + 2b\cdot\nabla\chi + \rho\Delta\chi$, its total variation is at most CR^{d-1}. Integration by parts is justified by uniform and local H^1 mollification of the continuous function v. Bounding the right side by $C\delta R^{d-1}$ and splitting the square over A and $D \setminus A$ proves the estimate. Finally $|b| \geq c$ almost everywhere, by Euler's identity and $\rho(x) \geq c|x|$. □

Corollary 5.2 (Exact rigidity)**.** *If $U_D \geq 0$ and $U_D = 0$ on D^c, then, up to null sets, $D = rK$ for some $r \geq 0$ and $U_D = 2d\kappa(r-\rho)_+$.*

Proof. If $|D| = 0$, take $r = 0$. Otherwise take $\delta = 0$. Then $D = \{u > 0\}$ up to null sets and $\nabla u = -a\mathbf{1}_{\{u>0\}}\nabla\rho$. Thus u is Lipschitz and $u + a\rho$ is constant on each connected component of its positivity set. On such a component write $u = a(r - \rho)$. The segment from any point of the component toward the origin cannot leave it: at a first exit its value would still be strictly positive. Every component therefore contains the origin, so there is only one. Its boundary lies in $\{\rho = r\}$; following each radial segment now shows that the component is rK. □

Corollary 5.3 (Few sites have small local time)**.** *On the coarse event, let δ_n be a uniform bound for $|\widetilde{\ell}_n - U_{D_n}|$. Then simultaneously for every $t \geq 0$,*

$$\#\{x \in A_n : \ell_n(x) \leq t\} \leq C(t + \delta_n)r_n^{d-1}.$$

Proof. Apply the lemma with truncation threshold $t + \delta_n$ and $R = Cr_n$. Every cell with $\ell_n(x) \leq t$ lies in $\{U_{D_n} \leq t + \delta_n\}$. □

6. Limit laws

Proof of Theorem 1.1(iv). Put $b(x) = \kappa\lambda(x)\xi(x)$. Euler's identity gives $x \cdot b(x) = \kappa\lambda(x)\rho(x)$. With I_j indicating a first departure,

$$Q_n = |X_n|^2 - n + 2\kappa S_n, \qquad \Delta Q_j = 2X_j \cdot (\Delta X_j + I_j b(X_j))$$

is a martingale. Outside first departures its increment variance is $4|X_j|^2/d$. Since $|b(x)| \leq 1$,

$$\langle Q\rangle_n = \frac{4}{d}\sum_x \ell_n(x)|x|^2 + O\left(\sum_{x\in A_n}|x|^2\right).$$

The error is $O(r_n^{d+2})$. The cone and a Riemann sum give

$$\frac{\langle Q\rangle_n}{r_n^{d+3}} \longrightarrow 8\kappa\int_K (1-\rho(z))|z|^2\,dz = \frac{8\kappa J_K}{d+3} \quad \text{almost surely.}$$

The last identity is homogeneity. The bound $|\Delta Q_j| \leq 4|X_j|$ is predictable and verifies the martingale Lindeberg and Stout increment conditions: its ratio to $\sqrt{\langle Q\rangle_n/\log\log\langle Q\rangle_n}$ is $O(\sqrt{\log\log n/n})$. Apply the martingale central limit theorem [3, Corollary 3.1] and iterated-logarithm law [7]. Since $|X_n|^2 = O(r_n^2)$ is negligible for $d \geq 2$, divide by 2κ to obtain the assertions for S_n. These imply $S_n \sim n/(2\kappa)$ almost surely, and differentiation gives the factor $1/(dV)$ for $R_{\mathrm{mom}}(n)$. □

Theorem 6.1 (Local times centered by the cookie potential)**.** *Under the assumptions of Theorem 1.1, use the Green kernel g of Lemma 2.2. Define the range-dependent potential and normalization by*

$$P_n(y) = \sum_{x\in A_n}\sum_{|e|=1}\left(\frac{1}{2d} - q_x(e)\right)g(x+e-y), \qquad s_n = \begin{cases}\sqrt{r_n\log r_n}, & d=2,\\ \sqrt{r_n}, & d\geq 3.\end{cases}$$

For every finite list $y_1,\dots,y_k$, $((\ell_n(y_i) - P_n(y_i))/s_n)_i$ tends to a centered Gaussian vector of covariance

$$\Sigma_{ij} = \begin{cases}16\kappa/\pi, & d=2,\\ 2d\kappa(2G(y_i-y_j) - \mathbf{1}_{\{y_i=y_j\}}), & d\geq 3.\end{cases}$$

Write $\Sigma(y,y)$ for the corresponding diagonal variance. At each fixed site, for either sign, almost surely,

$$\limsup_n \frac{\pm(\ell_n(y) - P_n(y))}{s_n\sqrt{2\log\log n}} = \sqrt{\Sigma(y,y)}.$$

Both assertions also hold with $U_{D_n}(y)$ in place of $P_n(y)$. A deterministic cone requires the centering estimate below.

Proof. The Dynkin martingale M^y for $g_y(x) = g(x-y)$ satisfies

$$\ell_n(y) - P_n(y) = -M_n^y + g(X_n - y) - g(-y).$$

The endpoint is $O(\log n)$ in $d = 2$ and $O(1)$ in $d \geq 3$. Define

$$\Gamma(f,h)(x) = \frac{1}{2d} \sum_{|e|=1} (f(x+e) - f(x))(h(x+e) - h(x)) - \Delta f(x)\Delta h(x).$$

The bound $|g(x+e) - g(x)| \leq C(1+|x|)^{1-d}$ shows that first-cookie covariance corrections total $O_{y,z}(\log r_n)$ in the plane and $O_{y,z}(1)$ in higher dimensions. Thus

$$\langle M^y, M^z \rangle_n = \sum_x \ell_n(x)\Gamma(g_y, g_z)(x) + O_{y,z}(1 + \log r_n).$$

For $d \geq 3$, summability, the cone profile and summation by parts give $r_n^{-1}\langle M^y, M^z\rangle_n \to 2d\kappa(2G(y-z) - \mathbf{1}_{\{y=z\}})$. In $d = 2$, the covariance kernel is $2/(\pi^2|x|^2) + O_{y,z}(|x|^{-3})$, whose sum over a ball of radius r is $(4/\pi)\log r + O(1)$. The constant $4\kappa r_n$ of the cone gives $(16\kappa/\pi)r_n \log r_n$; the term involving $\rho(x)$ and the change from a Euclidean ball to a gauge ball cost only $O(r_n)$. The profile error costs $o(r_n \log r_n)$. These are the bracket limits. Bounded increments allow the multivariate martingale CLT and the scalar LIL. The potential representation gives $P_n(y) = U_{D_n}(y) + O(\log n)$ eventually almost surely, which proves the assertion about continuum centering. □

Lemma 6.2 (Centering at a fixed site with sparse weights)**.** *Assume the Euclidean radius and volume estimates. Let U_D denote the* unweighted *radial continuum potential, and set*

$$q_n = \begin{cases} (\log n/r_n)^{1/2}, & d = 2, \\ \log n/r_n, & d \geq 3, \end{cases} \qquad w_n = \begin{cases} \sqrt{r_n}(\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \geq 3. \end{cases}$$

For every fixed y, almost surely for all large n,

$$U_{D_n}(y) - 2d\kappa(r_n - |y|) - \frac{Q_n}{\omega_d r_n^d} = O_y(1 + q_n w_n) = \begin{cases} O_y((\log n)^3), & d = 2, \\ O_y(1), & d \geq 3. \end{cases}$$

Consequently the actual local time satisfies

$$\ell_n(y) - 2d\kappa(r_n - |y|) = -M_n^{g_y} + \frac{Q_n}{\omega_d r_n^d} + \begin{cases} O_y((\log n)^3), & d = 2, \\ O_y(\log n), & d \geq 3. \end{cases}$$

Proof. Write $\nu = 1_{D_n} - 1_{B(0,r_n)}$. Its support lies in $\{v : ||v| - r_n| \leq Cw_n\}$ and $\|\nu\|_1 \leq Cr_n^d q_n$. On this annulus,

$$\left| |v|^{1-d} - r_n^{-d}|v| \right| \leq Cr_n^{-d}||v| - r_n|.$$

Integration against ν gives

$$U_{D_n}(0) - 2d\kappa r_n = \frac{2\kappa}{\omega_d r_n^d}\left(\int_{D_n} |v|\, dv - \frac{n}{2\kappa}\right) + O(q_n w_n).$$

Cell replacement costs $O(r_n^d)$. The sparse-weight condition gives

$$\left| S_n - \sum_{x \in A_n} |x| \right| \leq Cr_n \sum_{|x| \leq Cr_n} |\lambda_x - 1| = O(r_n^d).$$

Thus the parenthesis is $(Q_n - |X_n|^2)/(2\kappa) + O(r_n^d)$. Since $|X_n|^2/r_n^d = O(1)$, this proves the assertion at the origin. For fixed y the difference between the kernels at y and at zero is $O_y(r_n^{-d})$ on the support of ν, so its integral is $O_y(1)$. The ball potential changes by exactly $-2d\kappa|y|$. The local-time assertion follows from the Green-martingale representation with its uniform $O(\log n)$ error. □

Deterministic centering in the Euclidean case. The preceding lemma completes Theorem 1.2(ii) from Theorem 6.1. Indeed, the moment LIL gives

$$Q_n/r_n^d = O\big(r_n^{(3-d)/2}\sqrt{\log\log n}\big) \quad \text{almost surely,}$$

which, together with the centering error, is negligible on both the fixed-site CLT and LIL scales. The same Green martingales therefore give the asserted joint Gaussian law and signed iterated-logarithm laws.

7. Sharpness and fluctuation lower bounds

We continue with $\rho(x) = |x|$ and put $u_x = x/|x|$ for $x \neq 0$, $u_0 = 0$. Write $r = r_n$, $L = \log n$, $R_- = R_{\mathrm{in}}(n)$, and $R_+ = R_{\mathrm{out}}(n)$. All estimates below allow arbitrary first-cookie covariances. We use the already proved quantitative shape estimates on an event Ω_n, for a fixed constant $K_0 < \infty$, with $\mathbb{P}(\Omega_n^c) \le Cn^{-10}$:

$$R_+ \le K_0 r, \qquad \sup_x |\ell_n(x) - 2d\kappa(r - |x|)_+| \le Cr^{1/2}L^{3/2}, \tag{7.1}$$

$$|R_- - r| + r^{1-d}|D_n \triangle B_r| \le Ch_d, \qquad h_d = \begin{cases} \sqrt{rL}, & d = 2, \\ L, & d \ge 3. \end{cases} \tag{7.2}$$

The discrete-to-continuum potential estimate and Dynkin's identity then give, uniformly for $|y| \le r/2$,

$$\ell_n(y) - 2d\kappa(r - |y|) = -M_n^y + O(h_d + L) \quad \text{on } \Omega_n. \tag{7.3}$$

Indeed, $E = D_n \setminus B_{R_-}$ lies a distance of order r from such y, so its drift potential has absolute value at most $Cr^{1-d}|E| \le Ch_d$; the potential of B_{R_-} is the cone of height $2d\kappa R_-$, and the remaining error is $O(L)$.

7.1. One exponential-martingale lemma.

Lemma 7.1. *Fix $0 < c_0 \le C_0$. Let $S^1, \dots, S^m$ be real martingales starting at zero with increments bounded by b. Suppose an event Ω, of probability at least $1 - \alpha$, satisfies $c_0 \le \langle S^i\rangle_n \le C_0$ for every i. There are $c, C > 0$, depending only on c_0, C_0, such that if $\beta \ge C$ and $\beta b \le c$, then*

$$\mathbb{P}(\pm S_n^i \ge c\beta) \ge \tfrac{1}{4}e^{-C\beta^2} - \alpha.$$

If additionally $|\langle S^i, S^j\rangle_n| \le \delta$ on Ω for $i \neq j$ and $\beta^2\delta + \beta^3 b \le 1$, then

$$\mathbb{P}\{\max_i S_n^i < c\beta\} \le C\left(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3 b + e^{C\beta^2}\sqrt{\alpha}\right).$$

Proof. Stop all martingales just before any bracket would exceed $2C_0$; this changes none of them on Ω. For a martingale S whose increments are at most b_1 and whose bracket is bounded by $8C_0$, put

$$K_S(s) = \sum_{j<n} \log \mathbb{E}(e^{s\Delta S_j} \mid \mathcal{F}_j), \qquad Z_S(s) = e^{sS_n - K_S(s)}.$$

Conditional Taylor expansion, for $|s|b_1 \le 1/2$, gives

$$K_S(s) = \tfrac{1}{2}s^2\langle S\rangle_n + O(|s|^3 b_1), \quad \mathbb{E}Z_S(s) = 1, \quad \mathbb{E}Z_S(s)^2 \le e^{Cs^2}.$$

For the last bound use $Z_S(s)^2 = Z_S(2s)e^{K_S(2s)-2K_S(s)}$. Apply these formulas to S^i and $S^i + S^j$, with $s = \beta$, and abbreviate $Z_i = Z_{S^i}(\beta)$. On Ω, $Z_i \ge 1/2$ implies $S_n^i \ge c\beta$, since $K_{S^i}(\beta) \ge c_0\beta^2/4$. Paley–Zygmund therefore gives the first assertion; use $-S^i$ for the opposite sign. For $i \neq j$,

$$Z_iZ_j = Z_{S^i+S^j}(\beta)\exp\{\beta^2\langle S^i, S^j\rangle_n + O(\beta^3 b)\}.$$

On Ω the exponential is at most $1 + C(\beta^2\delta + \beta^3 b)$; on its complement apply Cauchy–Schwarz and the second-moment bound. Consequently

$$\mathbb{E}Z_iZ_j \le 1 + C(\beta^2\delta + \beta^3 b) + e^{C\beta^2}\sqrt{\alpha}.$$

The average $\overline{Z} = m^{-1}\sum_i Z_i$ has mean one and variance bounded by the right side of the asserted probability estimate. If $\overline{Z} \geq 1/2$ on Ω, some $S_n^i \geq c\beta$. Chebyshev, with the additional error α, finishes the proof. □

7.2. **Planar radii and width.** Let $d = 2$. Define the weighted moment and vector martingales

$$S_n = \sum_{x\in A_n} \lambda_x |x|, \qquad Q_n = |X_n|^2 - n + 2\kappa S_n, \qquad Z_n = X_n + \kappa \sum_{x\in A_n} \lambda_x u_x.$$

On $\{R_+ \leq K_0 r\}$ the defect assumption gives

$$\left| S_n - \sum_{x\in A_n} |x| \right| \leq Cr^2, \qquad \left| \sum_{x\in A_n} (\lambda_x - 1)u_x \right| \leq Cr. \tag{7.4}$$

The first-cookie covariance corrections contribute $O(r^4)$ to $\langle Q\rangle_n$ and $O(r^2)$ to each bracket of Z. Thus (7.1) gives, on Ω_n,

$$\langle Q\rangle_n = (4\pi\kappa/5 + o(1))r^5, \qquad \langle v\cdot Z\rangle_n = |v|^2 n/2 + O(|v|^2 r^2). \tag{7.5}$$

These formulas also hold almost surely with relative error $o(1)$ by the qualitative shape theorem.

The unweighted moment radius

$$T_n = \left(\frac{3}{2\pi} \sum_{x\in A_n} |x| \right)^{1/3}$$

satisfies $R_- \leq T_n + C$ and $R_+ \geq T_n - C$ by comparison with unit cells. The weighted moment radius differs from T_n by $O(1)$ on Ω_n and eventually almost surely, by (7.4) and the shape theorem. The general moment LIL therefore proves

$$\limsup_n \frac{r - R_-}{\sqrt{r\log\log n}}, \quad \limsup_n \frac{R_+ - r}{\sqrt{r\log\log n}} \geq \frac{1}{\sqrt{10\pi\kappa}} \quad \text{almost surely.} \tag{7.6}$$

To obtain polynomial lower tails, stop Q on exiting $B_{K_0 r}$ and apply Lemma 7.1 to $Q/r^{5/2}$. Its increments are bounded by $4K_0 r^{-3/2}$, and its bracket lies between positive constants on Ω_n by (7.5). Choose $\beta = a_p\sqrt{\log n}$, with $a_p > 0$ small enough that $Ca_p^2 < \min(p, 10)/2$. Each sign then has probability exceeding n^{-p} after intersecting with Ω_n and subtracting its failure probability once more, for large n. Since $|X_n|^2$ and the defect in (7.4) are $O(r^2) = o(r^{5/2}\sqrt{\log n})$, the identity for Q and factoring $T_n^3 - r^3$ yield, with a suitable $c_p > 0$,

$$\mathbb{P}\{r - R_- \geq c_p\sqrt{r\log n}\} \geq n^{-p}, \qquad \mathbb{P}\{R_+ - r \geq c_p\sqrt{r\log n}\} \geq n^{-p}. \tag{7.7}$$

For the width set $V_n = \sum_{x\in A_n} u_x$. The sum over the lattice disk $\{|x| < R_-\}$ vanishes by symmetry. When $R_-, R_+ \asymp r$, integration of the positive part of any directional component over the remaining annulus gives

$$|V_n| \leq R_+^2 - R_-^2 + Cr, \qquad |Z_n| \leq \kappa(R_+^2 - R_-^2) + Cr. \tag{7.8}$$

Indeed, $\int_0^{2\pi} (\cos\theta)_+ \, d\theta = 2$; replacing each unit vector by its values in the corresponding cell costs $O(r)$, as does the weighted correction. The martingale $e_1 \cdot Z$ has increments at most two. Apply Lemma 7.1 to $(e_1 \cdot Z)/\sqrt{n}$, using its bracket bounds on Ω_n from (7.5). Equation (7.8) and $n = (4\pi\kappa/3)r^3$ give

$$\mathbb{P}\{R_+ - R_- \geq c_p\sqrt{r\log n}\} \geq n^{-p}. \tag{7.9}$$

The same bracket limit, the bounded-increment martingale LIL, and $R_- + R_+ \sim 2r$ give

$$\limsup_n \frac{R_+ - R_-}{\sqrt{r\log\log n}} \geq \sqrt{\frac{\pi}{3\kappa}} \quad \text{almost surely.} \tag{7.10}$$

Finally the vector martingale CLT yields $Z_n/\sqrt{n} \Longrightarrow N(0, I_2/2)$. On $\{R_+ - R_- \le a\sqrt{r}\}$, (7.8) gives $|Z_n|/\sqrt{n} \le \sqrt{3\kappa/\pi}\, a + o(1)$ almost surely. Since $\mathbb{P}\{|N(0, I_2/2)| \le t\} = 1 - e^{-t^2}$,

$$\limsup_n \mathbb{P}\{R_+ - R_- \le a\sqrt{r}\} \le 1 - e^{-3\kappa a^2/\pi} \qquad (a \ge 0). \tag{7.11}$$

7.3. **The maximum of the bulk local times.** Let $m = \lfloor r^{1/8} \rfloor$, $y_i = i\lceil r^{1/4} \rceil e_1$, and put

$$f_i = \begin{cases} g_{y_i + k e_1} - g_{y_i}, & k = \lfloor r^{1/8} \rfloor, \quad d = 2, \\ g_{y_i}, & d \ge 3, \end{cases} \qquad s = \begin{cases} \sqrt{r \log r}, & d = 2, \\ \sqrt{r}, & d \ge 3, \end{cases} \qquad S^i = -M^{f_i}/s.$$

All the poles lie in $B_{\sqrt{r}}$ for large n. We verify, uniformly on Ω_n,

$$c_0 \le \langle S^i \rangle_n \le C_0, \qquad |\langle S^i, S^j \rangle_n| \le C r^{-1/4} \quad (i \ne j). \tag{7.12}$$

This calculation uses only the simple-random-walk covariance kernel Γ and the cone estimate; no independence is claimed. Write $W_{ij} = \sum_x |\Gamma(f_i, f_j)(x)|$ and $T_{ij} = \sum_x \min(|x|, r)|\Gamma(f_i, f_j)(x)|$. The Green-kernel gradient bound and $2ab \le a^2 + b^2$ give

$$(W_{ij}, T_{ij}) \le \begin{cases} (C \log r, C\sqrt{r} \log r), & d = 2, \\ (C, C(\sqrt{r} + \log r)), & d \ge 3. \end{cases}$$

For completeness, in $d \ge 3$ the squared increments are bounded by $C(1 + |x - y_i|)^{2-2d}$; its sum is finite and its first truncated moment is $O(\log r)$ in $d = 3$ and $O(1)$ in larger dimensions. In $d = 2$, an increment of f_i is bounded by the sum of reciprocal distances to its two poles within distance $3k$, and by $Ck(1 + |x - y_i|)^{-2}$ outside. Squaring and summing gives $O(\log r)$ and a first moment $O(k \log r)$, respectively. Translation of the poles costs at most $C\sqrt{r}$ times the unweighted sums. The finitely supported Laplacian terms in Γ satisfy the same bounds.

Replacing first-cookie covariances by Γ costs $O(1)$ in $d \ge 3$ and $O(\log r)$ in $d = 2$. Replacing ℓ_n by its cone and then by the constant $2d\kappa r$ costs at most $C r^{1/2} L^{3/2} W_{ij} + 2d\kappa T_{ij}$. After division by s^2 all these errors are $O(r^{-1/2} L^{3/2}) = o(r^{-1/4})$. Thus the bracket differs by this error from

$$\Sigma_{ij} = \frac{2d\kappa r}{s^2} \sum_x \Gamma(f_i, f_j)(x).$$

For $d \ge 3$, summation by parts gives $\Sigma_{ij} = 2d\kappa(2G(y_i - y_j) - \mathbf{1}_{\{i=j\}})$. Its diagonal is positive, and its off-diagonal entries are $O(r^{(2-d)/4})$. For $d = 2$, writing $z = y_i - y_j$, the same identity gives

$$\sum_x \Gamma(f_i, f_j)(x) = 2\{g(z + k e_1) + g(z - k e_1) - 2g(z)\} - 2\mathbf{1}_{\{i=j\}}.$$

The diagonal of Σ tends to $4\kappa/\pi$, because $g(k e_1) = (2/\pi) \log k + O(1)$. For $i \ne j$, the displayed second difference is $O(k^2/|z|^2)$, giving an off-diagonal bound $O(r^{-1/4})$. This proves (7.12).

The normalized martingale increments are bounded by $C r^{-1/2}$. Apply Lemma 7.1 with $\beta = a\sqrt{\log r}$, where $a > 0$ is so small that $Ca^2 < 1/16$. Since $m \asymp r^{1/8}$, $\delta = C r^{-1/4}$, and $\alpha = O(n^{-10})$, the probability that all $S_n^i < c\sqrt{\log r}$ is at most $C r^{-1/16}$. The error in (7.3) is $o(s\sqrt{\log r})$. In $d \ge 3$ a large S_n^i gives a large positive local-time deviation at y_i; in $d = 2$ it gives a large difference of the deviations at its two poles, so one has large absolute value. Consequently, for some $c > 0$,

$$\mathbb{P}\left\{ \max_{|y| \le \sqrt{r}} |\ell_n(y) - 2d\kappa(r - |y|)| \ge c \begin{cases} \sqrt{r} \log n, & d = 2, \\ \sqrt{r \log n}, & d \ge 3 \end{cases} \right\} \ge 1 - n^{-c} \tag{7.13}$$

for all sufficiently large n.

7.4. **Logarithmic radii under ellipticity.** For this paragraph assume additionally $q_x(e) \ge p_0 > 0$ for every x, e; later departures already have probability $1/(2d)$. There are $c, C, h_0 > 0$ such that, for $h_0 \le h \le r/8$,

$$\mathbb{P}\{R_- \ge r - h,\ R_+ \le r + h\} \le e^{-cr^{d-1}e^{-Ch}}, \tag{7.14}$$

$$\mathbb{P}\{R_+ \le r + h\} \le e^{-cr^{d-1}e^{-Ch}} + e^{-cr^{d-3}h^2}. \tag{7.15}$$

To prove these estimates, from any x with $|x| > r - 2h$ take $T = \lceil 4h\sqrt{d} \rceil$ steps in the signed coordinate of largest $|x_i|$. The Euclidean distance increases at speed at least $1/\sqrt{d}$, so this path exits B_{r+h}; its conditional probability is at least p_0^T after decreasing p_0 if necessary. Among k distinct arrivals at sites of any fixed $\Lambda \subset \{|x| > r - 2h\}$ by time n, at least $\lfloor (k - T)/(T + 1) \rfloor$ disjoint such trials can be started and completed by time n. Successive conditioning bounds their joint failure probability by $(1 - p_0^T)^{\lfloor (k-T)/(T+1) \rfloor}$. If $R_- \ge r - h$, the annulus $r - 2h < |x| < r - h - 1$ supplies $k \ge cr^{d-1}h$ sites, proving (7.14).

For (7.15), let $\Lambda = \{r - 2h < |x| \le r + h\}$. The same trials apply if $|A_n \cap \Lambda| \ge c_1 r^{d-1}h$. Otherwise, on $R_+ \le r + h$, cell comparison and the defect bound give

$$\sum_{x \in A_n} \lambda_x |x| \le \frac{n}{2\kappa} - c_2 r^d h + 2r|A_n \cap \Lambda| + Cr^d.$$

Choose c_1 small and then h_0 large. The identity $Q_n = |X_n|^2 - n + 2\kappa \sum \lambda_x |x|$ forces $Q_n \le -cr^d h$; the endpoint $|X_n|^2$ is absorbed by h_0 in $d = 2$ and by large r in higher dimensions. Stop Q on leaving B_{r+h}. Its increments are bounded by $4(r + h)$, so Azuma's inequality gives $\mathbb{P}\{Q_n \le -cr^d h,\ R_+ \le r + h\} \le e^{-cr^{d-3}h^2}$, proving the second estimate. Taking $h = c \log n$ with $c > 0$ sufficiently small and applying Borel–Cantelli gives, almost surely for all sufficiently large n,

$$\max(r - R_-, R_+ - r) > c \log n; \qquad R_+ - r > c \log n \quad \text{if } d \ge 3.$$

The ellipticity assumption is used only in the prescribed-path trials and is not needed for any preceding lower bound.

Draft and verification scope. This is a new mathematical draft. The Lean certificate accompanying [1] is pinned to that paper and does not certify the generalized statements here.

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